

Test taxonomy

Not every test should be the same shape

Four tiers

- Directed means hand-crafted stimulus for a known scenario
- Random means constrained-legal traffic to explore space
- Stress means long-run, backlog, or concurrent pressure
- Corner means illegal, framing, or rare edges
- Open rows are untyped

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Test taxonomy planner

Document-aid literacy: classify tests into **directed**, **random**, **stress**, and **corner** tiers.

Starter: one test per tier — plan BALANCED.

Starter example: uart_byte→directed, uart_stream→random, fifo_soak→stress, parity_bad→corner — plan BALANCED.

Load starter example

Challenges 0 / 22 cleared

Quiz: directed: A directed test is...

☐ hand-crafted stimulus for a known scenario or requirement ☐ only illegal opcodes ☐ a Makefile PHONY target

☐ always a multi-day soak

Show hint Check Next Idle

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22

Core ideas

directed

Known scenario / requirement depth.

random

Constrained-random legal breadth.

stress

Long-run / backlog pressure.

corner

Illegal / boundary / negative.

Lab

SCENARIO

Lab starter

Planning docs practice

- List four short test names for a UART-like block and tag each directed
- If everything is directed, rewrite one as random and one as corner
- Say in one sentence why the mix reduces escape risk

Pitfalls to watch

- Do not equate balanced taxonomy with one-hundred percent functional coverage
- Do not leave tests untyped on a living plan
- Do not hide negative cases because they are “hard.” And do not confuse a green CI job with

Your turn

- Complete the checklist for at least one track, preferably both
- Classify a small set until it looks balanced